RM9 continuation: priority allocation, control effects and conservative lane-flow readiness

Method-first ramp-metering review

1 October 2026

Disposition. RM9-UG1.A is approved as **Default**. The selected merge subproblem is transcribed below. The complete lane-flow coupling is not adopted; Section 4 is an explicit proposal for RM9-UG2. No traffic simulation, numerical derivative check or useful-gradient certificate is reported. Proofs and examples are reviewer derivations.

1 The approved two-input allocator

Let $d_m, d_r \geq 0$ be local mainline and ramp sending offers and $S \geq 0$ their common receiving rate, all in vehicle equivalents per unit time. Let structural priorities $p_m, p_r > 0$ be independent of offers and meter rate, with $a = p_m/(p_m + p_r) \in (0, 1)$. A transcription of [1]'s MISO law is

$$r = \min\{d_r, \max\{(1-a)S, S - d_m\}\}, \tag{1}$$

$$m = \min\{d_m, S - r\}. \tag{2}$$

The max term lets the ramp use supply the mainline cannot consume; the second line lets the mainline consume supply the ramp cannot consume. It is equivalent to the symmetric expression $m = \min\{d_m, \max\{aS, S - d_r\}\}$. This is accepted physical flow, not an optimized assignment or guaranteed entry.

Bounds and work conservation. The inner max lies in $[0, S]$, hence $0 \leq r \leq \min(d_r, S)$ and $0 \leq m \leq d_m$, $m + r \leq S$. If $d_m + d_r \leq S$, then $S - d_m \geq d_r$, giving $r = d_r, m = d_m$. If $d_m + d_r > S$ and $d_m \geq aS$, then $r = \min(d_r, (1-a)S)$: when the ramp is under its share, $m = S - d_r$; otherwise $m = aS$. If $d_m < aS$, then $r = S - d_m, m = d_m$. Consequently

$$m + r = \min\{S, d_m + d_r\}. \tag{3}$$

This proof covers zero supply/offers. No source-wide MIMO/FIFO behavior follows.

Ordinary derivatives, not a verified whole-chain gradient. At fixed d_m, S, a, write $B = \max((1-a)S, S - d_m)$. Away from $d_r = B$,

$$\frac{\partial r}{\partial d_r} = \begin{cases} 1, & d_r < B, \\ 0, & d_r > B, \end{cases} \qquad \frac{\partial (m+r)}{\partial d_r} = \begin{cases} 1, & d_m + d_r < S, \\ 0, & d_m + d_r > S. \end{cases} \tag{4}$$

At ties the ordinary derivative need not exist; no smoothing or generalized estimator is adopted. In a supply-limited region, increasing ramp service can reduce mainline service one for one. This does not prove a useful meter gradient.

2 What the rule changes in the control chain

Hand-calculated service examples

Use $S = 180$ veh-equivalents/h and $a = 2/3$ solely for illustration. The nominal mainline/ramp shares are 120/60. No site calibration is implied.

d_m	d_r	m	r	Total / interpretation
180	30	150	30	180; mainline uses spare ramp share
180	10	170	10	180; allocation changes, total unchanged
180	80	120	60	180; 20 of ramp offer remains unserved
80	70	80	70	150; all eligible offers served
80	40	80	40	120; less offer leaves more idle supply

Rows compare local offers, not instantaneous upstream meter values. An upstream rate $u(t)$ bounds release $e(t)$; stored/traveling ramp traffic determines $d_r(t)$ later. A co-located meter is a separate geometric specialization. Post-meter traffic and blocked/unserved demand must remain in accounting.

Spare supply on another lane is not freely transferable

For two separate receivers of supply 180 each, ignore lateral access at the instant considered. Offers $(d_{m1}, d_{m2}, d_r) = (180, 50, 30)$ give

$$F = \min(180, d_{m1} + d_r) + \min(180, d_{m2}) = 180 + 50 = 230. \tag{5}$$

Offers $(90, 140, 30)$ have the same total offer 260 but give $F = 120 + 140 = 260$. Thus lane-specific eligibility can matter to total service. These are offer-level examples, **not matched density/flow/speed histories**, feasible controlled trajectories, or a proof that lane contrast changes the best meter action. A physical redistribution law is needed to connect them to traffic dynamics.

Why queue redistribution may produce no objective benefit

With all affected uncompleted mass M, common arrivals A, final system exit rate E, and common clearance horizon T_c,

$$\dot{M} = A - E, \qquad J = \int_0^{T_c} M(t)\, dt, \qquad M(T_c) = 0. \tag{6}$$

Identical initial total mass, arrivals and exit history imply identical J, even if ramp and mainline queues differ. The local merge flow $m + r$ is not necessarily E: downstream travel, storage and bottlenecks must be included. On a fixed differentiable branch, for common forcing and a control parameter b,

$$\frac{dJ}{db} = -\int_0^{T_c} (T_c - t) \frac{\partial E(t; b)}{\partial b}\, dt. \tag{7}$$

Allocation changes next lane/ramp masses and may change later receiving, spillback, lane access or exit timing. Those are possible causal paths, not established effects in this task. If final departure history stays unchanged, queue relocation alone gives zero total-time improvement.

3 Why the selected allocator does not close the lane model

A source coefficient range diagnostic

The fuller source [2], equation 6, uses a coefficient whose local numerator contains a combined incentive. Let $x > 0, y \geq 0$ denote donor and target weighted densities on a constant profile, and denote that coefficient by P. From the printed expression, reviewer algebra gives

$$P > 1 \quad \Longleftrightarrow \quad (I-1)x > 2y \tag{8}$$

on its positive branch. For the printed free-flow cooperation incentive near a merge, with base incentive one and other incentives inactive, $I = 2 + y/x$. Thus the isolated coefficient is $2x/(x+y)$, exceeding one if $x > y$. A coefficient called a fraction is therefore not automatically a donor-bounded accepted transfer. This diagnosis does *not* establish reachability of every such combination or failure of every complete source trajectory. Later operators or domain restrictions might resolve it; they must be shown, not assumed.

Zero-receiving special branch

For capacity $C > 0$, supply $S \in [0, C]$ and positive proposed lateral rate ϕ, source equations 16–17 give $\vartheta = 1 - S/C$ and the printed accepted rate

$$\lambda = \begin{cases} \phi, & \vartheta = 1, \\ (1-\vartheta)\phi, & \vartheta < 1. \end{cases} \tag{9}$$

At $S = 0$ this isolated operator accepts ϕ, whereas its limit from $S > 0$ is zero. It cannot be used as a simple receiving-bounded interface without an additional interpretation/domain condition. No branch is silently corrected. Source equations 20–22 also modify demand and supply after lateral transfer; applying another independent node operator could double-count budgets. This is an unresolved joint-update transcription, not a blanket source rejection.

An inspected predecessor does not supply the missing autonomy

Source [3]'s 2014 formulation optimizes lateral, longitudinal and ramp flows jointly. Its constraints are useful conservation precedents, but its lateral flow is not autonomous response to a meter-only action. The 2015 Part I article [4] has verified primary metadata; its full methods were not retrieved in this pass. A bounded attractiveness equation reproduced in [2], equation 5, is an inspected antecedent, not verification of [4]'s complete model.

What has and has not passed

Passed in this analytic review	Still required
Two-input allocation bounds, work conservation and fixed-branch derivatives	Complete conservative lateral/longitudinal/ramp update
Local service examples and full-time accounting identity	Exact observer/initialization and feasible identical-history pair
Source range/branch and actuator-scope diagnostics	Action relevance, full derivative correctness and independent usefulness

4 A concrete conservative adaptation for RM9-UG2.A

Proposed, not adopted. Use two physical mainline lanes per cell, with no instantaneous rest-group mixing. The update below is a reviewer adaptation; it is not the complete calibrated algorithm in [2] or [4]. It retains only discretionary density-driven transfer, ordinary first-order receiving and the Default ramp allocator. It adds no capacity-drop penalty.

For cell length L, timestep Δt, lane mass $n_\ell = Lk_\ell$, maximum mass $N_\ell = Lk_\ell^{\mathrm{jam}}$ and known positive parameters,

$$D_\ell(n) = \min(v_\ell n/L, C_\ell), \qquad S_\ell(n) = \min(C_\ell, w_\ell(N_\ell - n)/L). \tag{10}$$

Use the step restrictions $\Delta t v_\ell/L \le 1$ and $\Delta t w_\ell/L \le 1$. For $0 \le n \le N_\ell$ they imply $\Delta t D_\ell(n) \le n$ and $\Delta t S_\ell(n) \le N_\ell - n$. Parameters may differ between lanes; symmetry is not imposed to hide state.

Stage 1: bounded autonomous within-cell lateral exchange. For structural $\mu_{\ell j} \in [0,1]$, $\kappa_{\ell j} \in (0,1]$, use the bounded attractiveness form reproduced in [2], equation 5:

$$A_{\ell j} = \mu_{\ell j} \max\left(0, \frac{\kappa_{\ell j} k_\ell - k_j}{\kappa_{\ell j} k_\ell + k_j}\right), \tag{11}$$

$$\lambda_{\ell j} = \min(A_{\ell j} D_\ell(n_\ell), S_j(n_j)), \tag{12}$$

$$n_\ell^L = n_\ell + \Delta t(\lambda_{j\ell} - \lambda_{\ell j}). \tag{13}$$

At zero denominator set $A_{\ell j} = 0$ (both lane masses zero). Otherwise $0 \le A_{\ell j} \le 1$. There is at most one adjacent target per donor in the two-lane case. Both directions cannot be positive: that would require $k_j < \kappa_{\ell j} k_\ell$ and $k_\ell < \kappa_{j\ell} k_j$, contrary to $\kappa_{\ell j}\kappa_{j\ell} \le 1$. The bounds imply

$$0 \le n_\ell^L \le N_\ell, \qquad n_1^L + n_2^L = n_1 + n_2. \tag{14}$$

Rejected propensity does not leave the donor. For more lanes the same formula cannot be applied independently without a joint donor/receiver budget.

Stage 2: recompute state limits and retain unspent shared budgets. Recomputing from n^L alone could spend the original receiving or sending budget twice. Define residual longitudinal budgets for each cell-lane:

$$\overline{D}_{i\ell} = \min\{D_{i\ell}(n_{i\ell}^L), D_{i\ell}(n_{i\ell}) - \lambda_{\ell j}\}, \tag{15}$$

$$\overline{S}_{i\ell} = \min\{S_{i\ell}(n_{i\ell}^L), S_{i\ell}(n_{i\ell}) - \lambda_{j\ell}\}. \tag{16}$$

These residuals are nonnegative by Stage 1's bounds. At an ordinary interface use $q_{i\ell} = \min(\overline{D}_{i\ell}, \overline{S}_{i+1,\ell})$. At the shared shoulder receiver apply (1)–(2) with the mainline residual donor offer and receiver residual $\overline{S}$. With total accepted longitudinal/ramp incoming rate $f_{i\ell}^{\mathrm{in}}$,

$$n_{i\ell}^+ = n_{i\ell}^L + \Delta t(f_{i\ell}^{\mathrm{in}} - q_{i\ell}), \qquad f_{i\ell}^{\mathrm{in}} \le \overline{S}_{i\ell}, \quad q_{i\ell} \le \overline{D}_{i\ell}. \tag{17}$$

Thus incoming lateral plus longitudinal/ramp flows share the original S, and outgoing lateral plus longitudinal flows share the original D. Post-lateral limits also imply $0 \le n_{i\ell}^+ \le N_{i\ell}$. Every internal transfer cancels between its donor/receiver. Ramp/external waiting and post-meter propagation need corresponding conservative balances. This is an explicitly conservative shared-budget adaptation, not calibration, a selected geometry, a history certificate or verified useful gradient. Stage order and step dependence can restrict flows and must remain explicit.

5 Next decision and evidence order

RM9-UG2 asks whether to adopt the restricted candidate above for analytic review, or retain the fuller incentive/cooperation family and first resolve its source-level coupling. The expanded alternatives, pros, cons and dependencies are in `reporting/rm9_30_allocator_contract_and_control_chain.md`. The provisional recommendation is the restricted candidate, with no positive outcome assumed. It can legitimately establish aggregate sufficiency or no useful metering distinction. Fuller cooperation/capacity-loss mechanisms remain possible extensions; their necessity is not yet proved.

After adoption of a defensible law, transcribe the actual strong density, flow, speed, cumulative-count and ramp-history functions with temporal/spatial support, initialization, common forcing and known parameters. Check reconstruction before seeking an exact feasible hidden-history pair. A scalar equivalent of lane contrast remains admissible. Do not use equal speeds, weakened speed observations, hidden prehistory or an imposed service penalty to produce ambiguity.

Only after forward action relevance should the review claim a candidate sensitivity path. Ordinary branch derivatives are insufficient at switching boundaries, and they do not verify independent downstream usefulness. No finite-difference/autograd run, optimizer, SUMO experiment or successor is opened. RM9 remains open at the physical-law decision; RM9-UG1 stays closed.

Primary references and provenance

1. Wright et al. (2017), *On node models for high-dimensional road networks.* Existing RM0018, published author-hosted version, Section 3.1. DOI 10.1016/j.trb.2017.09.001.

2. Nagalur Subraveti, Knoop and van Arem, *First order multi-lane traffic flow model—an incentive based macroscopic model to represent lane change dynamics.* Existing FB0030, final PDF, printed pp. 1762–1766. Primary final PDF.

3. Roncoli, Papageorgiou and Papamichail (2014), *Optimal control for multi-lane motorways in presence of vehicle automation and communication systems.* Existing FB0034, IFAC proceedings pp. 4179–4181. Primary proceedings PDF.

4. Roncoli, Papageorgiou and Papamichail (2015), *Traffic flow optimisation in presence of vehicle automation and communication systems—Part I.* RM9-L01 **metadata lead only**; no full-method adoption. DOI 10.1016/j.trc.2015.06.014.

Ownership. The allocator transcription, analytical proofs, hand-calculated illustrations, range/branch diagnostics and proposed split update are reviewer work under their stated assumptions. They are not findings reported by all the cited papers or a verified combined literature gradient chain.